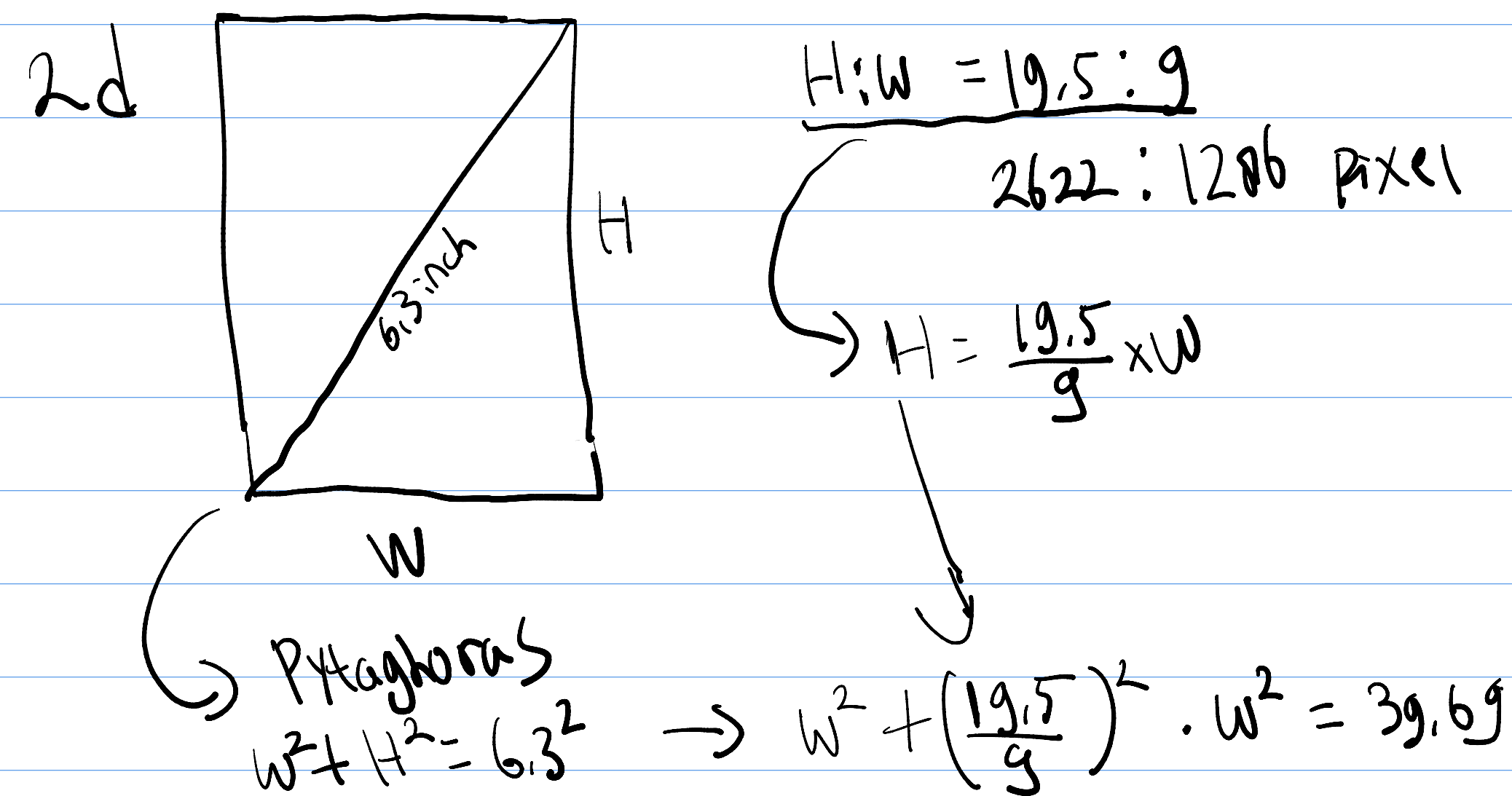


mmdb - Ex02

1. a. Vectorial = Uses mathematical formula to define paths, points and curves, allowing them to be infinitely scalable without loss of quality
Raster = Pixel based grids where each square holds a specific color, making them ideal for photographs
- b. GIF = 8 bit indexed, animation,
PNG = Lossless scaling, pre filtered, deflate
JPEG = Lossy DCT, 8x8 blocks, quantization table



$$W^2 \cdot (1 + 4.6944) = 39.69$$
$$\rightarrow W^2 = 39.69 / 5.6944 = 6.97$$
$$W = 2.64 \text{ inches}$$

$$\text{PPI (Pixel density)} = \frac{1206}{2.64} = 456 \text{ PPI}$$

3a Noise dithering is an white noise image intentionally added so limited-color images look less banded and more natural

- b. Uniform = - Choose how many colors for the final image
- Divide each color channel into equal sized images
- Assign each pixel to one of those ranges based on color values
- Replace each pixel with the representative color of its range

Median cut = - Put all image colors into one group
- Find the color channel with the widest spread of values
- Sort the colors in that group along the channel

- Split the group into two halves at the median point
- Repeat split process until desired color group emerges
- Compute one representative color for each group
- Replace each pixel with the representative color of the group it belongs to.